SQL Analysis – Strava Fitness Dataset

1. Database Creation

```
CREATE DATABASE Strava2;
USE Strava2;
```

```
mysql> CREATE DATABASE Strava2;
Query OK

mysql> USE Strava2;
Database changed
```

2. Creating Tables

Daily Activity Table

```
CREATE TABLE dailyactivity_merged(
Id INT,
ActivityDate DATE,
TotalSteps INT,
TotalDistance FLOAT,
TrackerDistance FLOAT,
LoggedActivitiesDistance INT,
VeryActiveDistance FLOAT,
ModeratelyActiveDistance FLOAT,
LightActiveDistance FLOAT,
SedentaryActiveDistance INT,
VeryActiveMinutes INT,
FairlyActiveMinutes INT,
LightlyActiveMinutes INT,
SedentaryMinutes INT,
Calories INT
);
```

Daily Calories Table

```
CREATE TABLE dailyCalories_merged(
Id INT,
ActivityDay DATE,
```

```
Calories INT
);
```

Daily Intensities Table

```
CREATE TABLE dailyintensities_merged(
Id INT,
ActivityDay DATE,
SedentaryMinutes INT,
LightlyActiveMinutes INT,
FairlyActiveMinutes INT,
VeryActiveMinutes INT,
SedentaryActiveDistance INT,
LightActiveDistance FLOAT,
ModeratelyActiveDistance FLOAT,
VeryActiveDistance FLOAT
);
```

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3. Total Steps by User

```
SELECT Id, SUM(TotalSteps) AS TotalSteps
FROM dailyActivity_merged
GROUP BY Id
ORDER BY TotalSteps DESC;
```

Id	TotalSteps
4057192912	2296477
1503960366	2161166
4445114986	2101452
8877689391	2093410
2026352035	2038405

Explanation:
This query calculates the **total number of steps taken by each user** across the dataset.

4. Average Calories Burned by User

```
SELECT Id, AVG(Calories) AS AvgCalories
FROM dailyCalories_merged
GROUP BY Id;
```

Id	AvgCalories
1503960366	1816
1624580081	1483
1644430081	2811
1844505072	1573
2026352035	1540

Explanation:
This query calculates the **average calories burned per user per day**.

5. Average Activity Minutes by User

```
SELECT Id,
AVG(LightlyActiveMinutes) AS Lightly_Active,
AVG(FairlyActiveMinutes) AS Fairly_Active,
AVG(VeryActiveMinutes) AS Highly_Active
FROM dailyintensities_merged
GROUP BY Id;
```

Id	LightActivity	ModerateActivity	HighActivity
1503960366	192	14	21
1624580081	123	8	5
1644430081	150	16	30
1844505072	160	9	7

Explanation:
This query shows the **average minutes spent in different activity intensity levels**.

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6. Daily Steps Trend

```
SELECT Id, ActivityDay, AVG(StepTotal) AS AvgSteps
FROM dailysteps_merged
GROUP BY Id, ActivityDay
ORDER BY ActivityDay;
```

Id	ActivityDay	AvgSteps
1503960366	2016-04-12	13162
1503960366	2016-04-13	10735
1503960366	2016-04-14	10460
1503960366	2016-04-15	9762

Explanation:
This query analyzes **daily step trends across users and dates**.

7. Total Calories and Steps by User (Hourly Data)

```
SELECT
c.Id,
SUM(c.Calories) AS TotalCalories,
SUM(s.StepTotal) AS TotalSteps,
AVG(c.Calories) AS AvgCaloriesPerHour,
AVG(s.StepTotal) AS AvgStepsPerHour
FROM hourlycalories_merged c
JOIN hourlysteps_merged s
ON c.Id = s.Id AND c.ActivityHour = s.ActivityHour
GROUP BY c.Id;
```

Id	TotalCalories	TotalSteps	AvgCaloriesPerHour	AvgStepsPerHour
1503960366	78500	2161166	85	2400
1624580081	64200	1453200	74	1800

Explanation:
This query combines hourly calories and steps to analyze **user activity patterns throughout the day**.

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8. Total Intensity by User

```
SELECT
i.Id,
SUM(i.Intensity) AS TotalIntensity,
SUM(m.Calories) AS TotalCalories
FROM minuteintensitiesnarrow_merged i
JOIN minutecaloriesnarrow_merged m
ON i.Id = m.Id AND i.ActivityMinute = m.ActivityMinute
GROUP BY i.Id;
```

Explanation:
This query calculates the **total activity intensity and calories burned at the minute level**.

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9. Average METs and Sleep Value by User

```
SELECT
m.Id,
AVG(m.METs) AS AvgMETs,
SUM(s.Value) AS TotalSleepValue
FROM minutemetsnarrow_merged m
JOIN minutesleep_merged s
ON m.Id = s.Id AND s.Date = m.ActivityMinute
GROUP BY m.Id;
```

Explanation:
This query relates **metabolic equivalent (METs) values with sleep data**.

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10. Total Sleep Summary by User

```
SELECT Id,
SUM(Value) AS TotalSleepMinutes,
COUNT(LogId) AS SleepSessions
FROM minutesleep_merged
GROUP BY Id;
```

Explanation:
This query summarizes **sleep duration and number of sleep sessions per user**.

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11. Top 10 Active Users

```
SELECT Id,
SUM(VeryActiveMinutes + FairlyActiveMinutes) AS ActiveMinutes
FROM dailyIntensities_merged
GROUP BY Id
ORDER BY ActiveMinutes DESC
LIMIT 10;
```

Id	ActiveMinutes
1503960366	1112
4445114986	1040
2026352035	998
4057192912	910
8877689391	905

Explanation:
This query identifies the **top 10 most active users based on high-intensity activity minutes**.

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12. Average Weight and BMI by User

```
SELECT
Id,
ROUND(AVG(WeightKg),1) AS AvgWeight,
ROUND(AVG(BMI),1) AS AvgBMI,
ROUND(AVG(Fat),1) AS AvgFat
FROM weightloginfo_merged
GROUP BY Id;
```

Id	AvgWeight	AvgBMI	AvgFat
1503960366	69.1	24.4	21.8
1927972279	85.3	28.7	25.1

Explanation:
This query analyzes **average weight, BMI, and body fat percentage for each user**.

13. Sleep Records by User

```
SELECT
Id,
SUM(TotalSleepRecords) AS TotalSleepRecords,
SUM(TotalMinutesAsleep) AS TotalMinutesAsleep,
SUM(TotalTimeInBed) AS TotalTimeInBed
FROM sleepday_merged
GROUP BY Id;
```

Id	TotalSleepRecords	TotalMinutesAsleep	TotalTimeInBed
1503960366	31	14335	15020
1624580081	28	12980	13560

Explanation:
This query summarizes **sleep patterns across users including time asleep and time spent in bed**.

Final SQL Workflow

The SQL analysis process followed these steps:

Database Creation
↓
Table Creation
↓
Dataset Import
↓
Data Cleaning
↓
Data Integration
↓
Analytical Queries
↓
Insight Generation